

FINAL PROJECT DEMONSTRATION

EEE 4518: ELECTRICAL AND ELECTRONIC WORKSHOP

PROJECT TITLE: ENHANCEMENT OF BATTERY-POWERED RICKSHAW

Submitted by – Team Murubb_EEE



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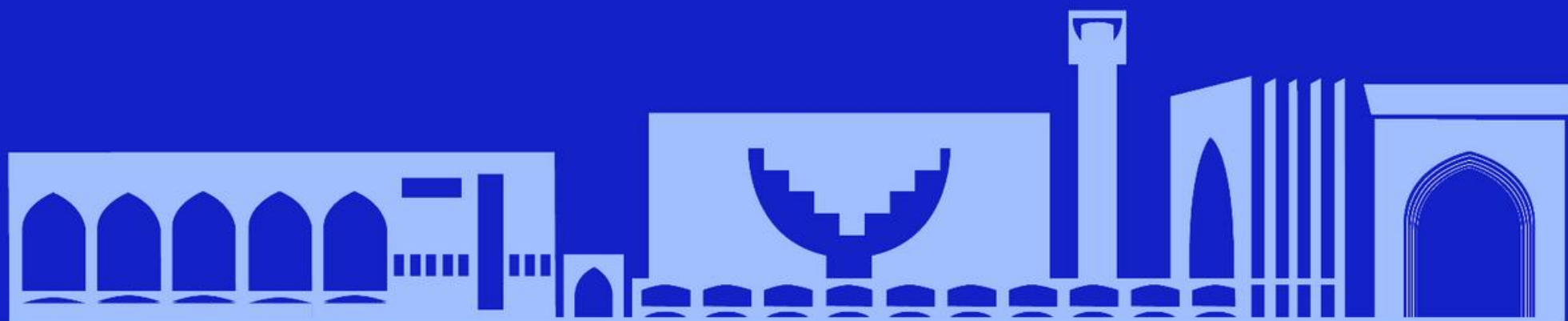
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OUTLINE

- 1.ABSTRACT**
- 2.PROBLEM STATEMENTS AND PROPOSED SOLUTIONS**
- 3.DESIGN**
- 4.IMPLEMENTATION**
- 5.ANALYSIS AND EVOLUTION**
- 6.REFLECTION OF INDIVIDUAL AND TEAMWORK**
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ABSTRACT

We, team Murubbeee, aim to work on enhancing battery-run rickshaws in Bangladesh. Battery rickshaws have become a common urban transportation in Bangladesh, especially in busy cities like Dhaka, and Chittagong. According to Dhaka Tribune's report on May 24, 2024, almost 60 lakh battery rickshaws are operating on the streets of Bangladesh. These rickshaws provide us with the benefits of fast transportation as compared to their counterpart, human-powered rickshaws. However, these benefits come with a price. Battery rickshaws are the major reasons for accidents, excessive electricity usage, and traffic-related issues. To address these issues and also upgrade these battery-run rickshaws for more efficient results, we propose our newly modified battery rickshaws.

PROBLEM STATEMENTS

1. Unnecessary speeding of battery rickshaws.
2. Collisions that lead to minor or severe injuries and life threats
3. Excessive use of electricity for charging the battery-run rickshaws
4. Lack of safety measurements in the rickshaws
5. Absence of an information system in the rickshaw.





PROPOSED SOLUTIONS

1. Solar panel installation on the top of rickshaw hoods
2. Speed sensors to detect the speed of the rickshaw
3. Obstacle detector to be installed at the front portion
4. A system that can respond to overspeed and reduce chances of collisions by controlling the power supply to motor.
5. Addition of indicator lamps and headlamps
6. Information system (LED display) to show necessary data (speed, charge, etc.)

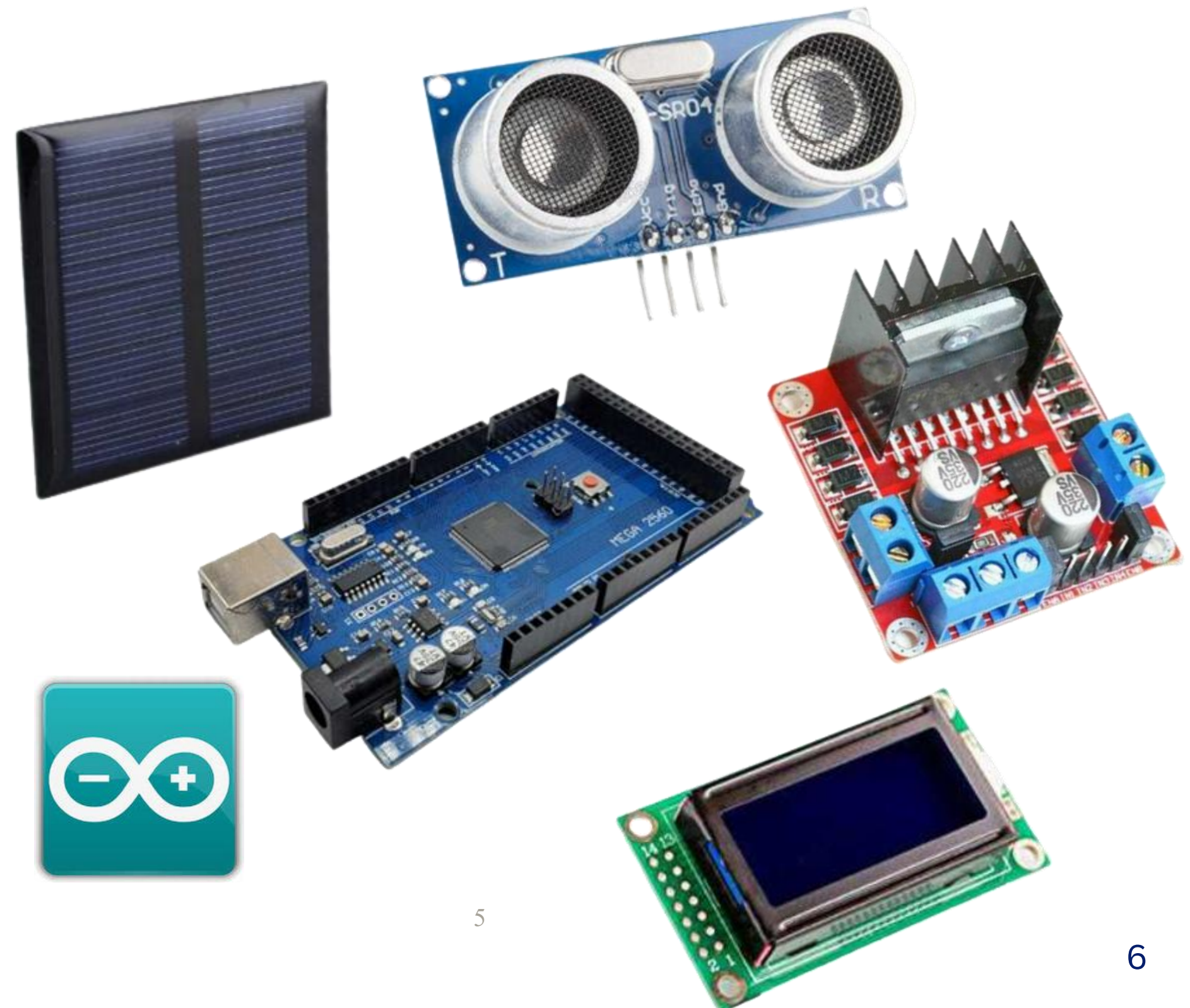
3.1 DESIGN: METHODS

Components Used in the prototype:

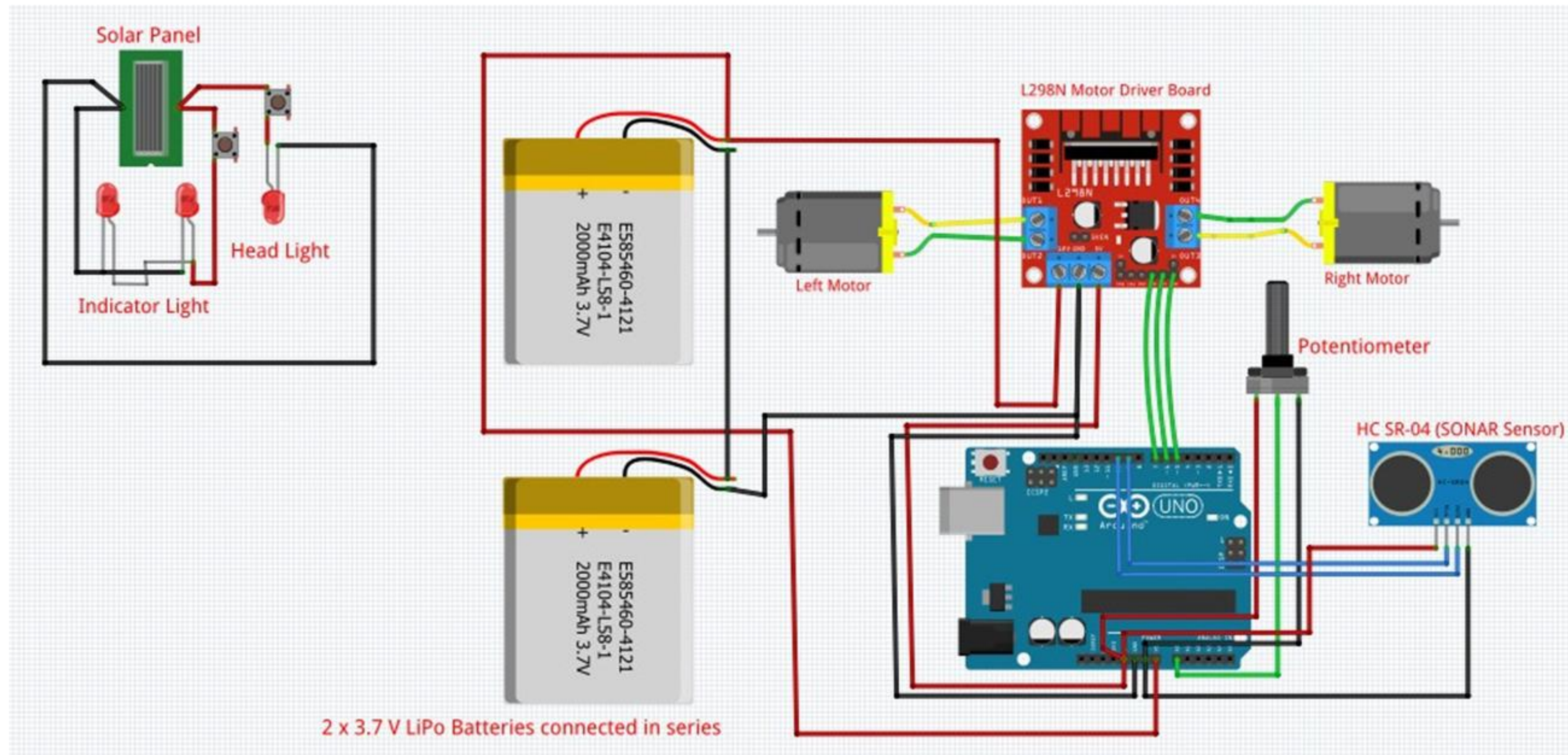
- ARDUINO
- 400 RPM motors
- Dual Motor Drivers
- Solar Panel
- Speed Measuring Module
- LCD Panels
- SONAR sensor
- LED indicators

Microcontroller Used: ATmega328

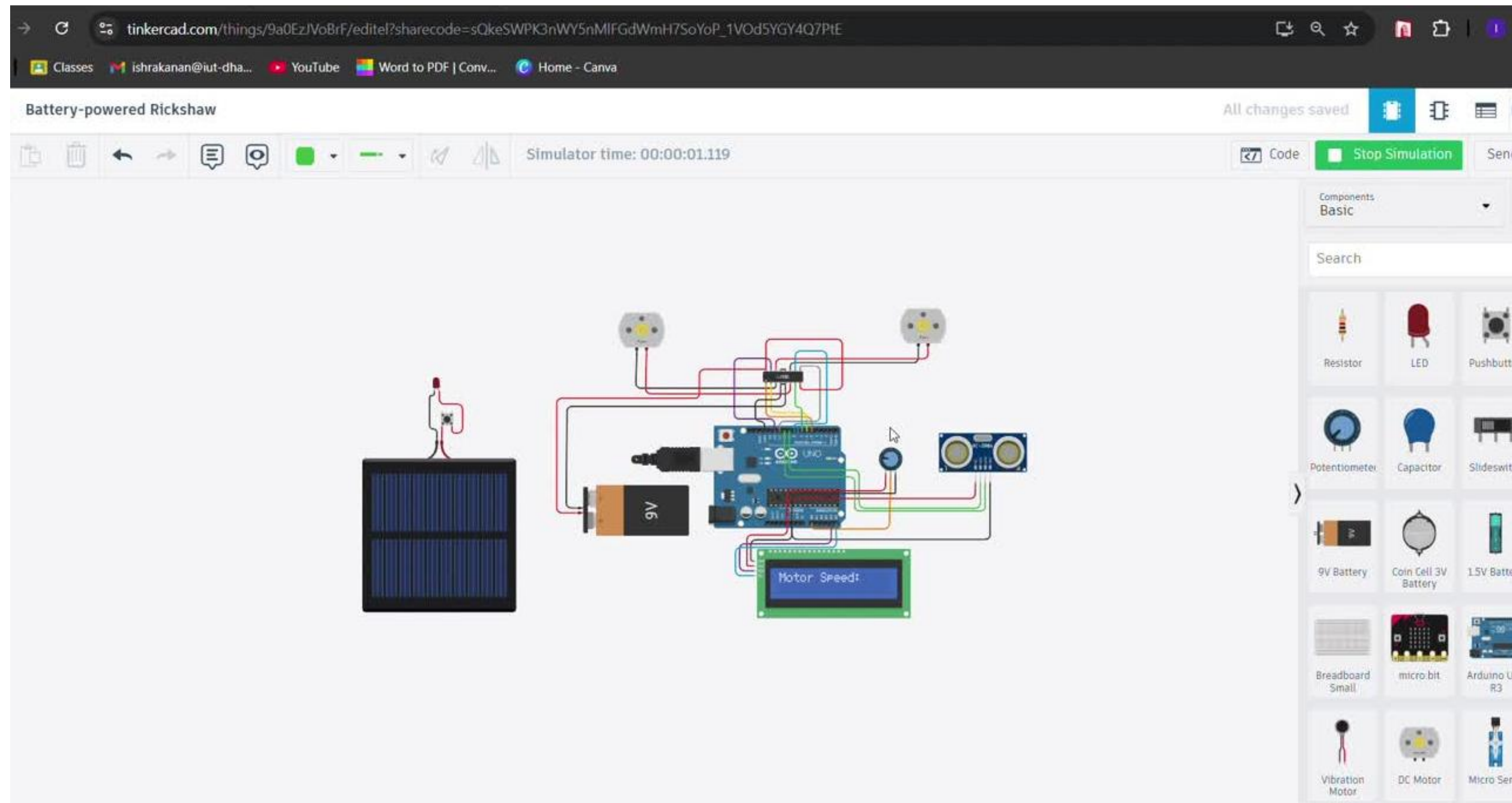
Code implemented by: Arduino IDE



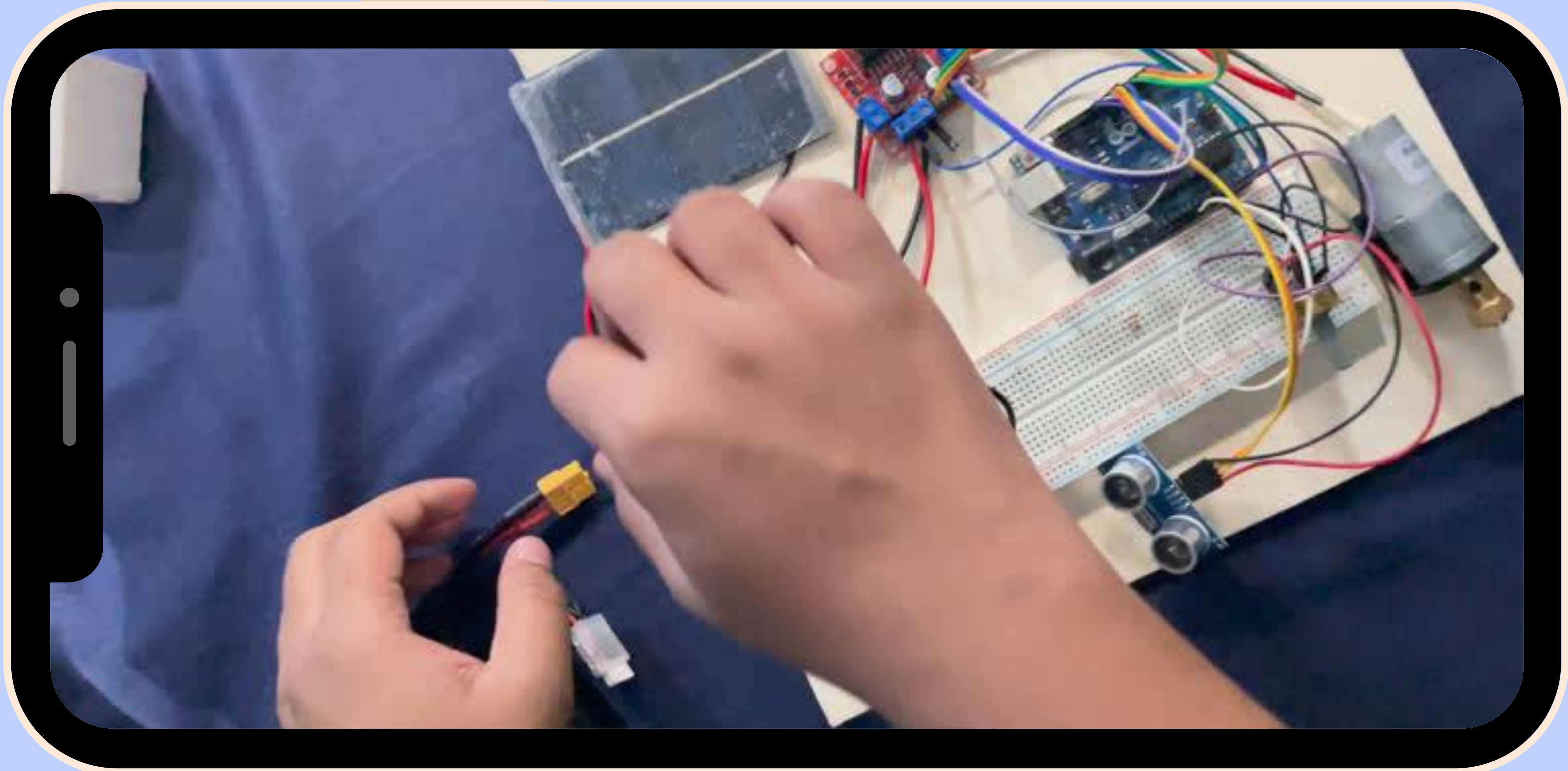
3.2 DESIGN: CIRCUIT DIAGRAM



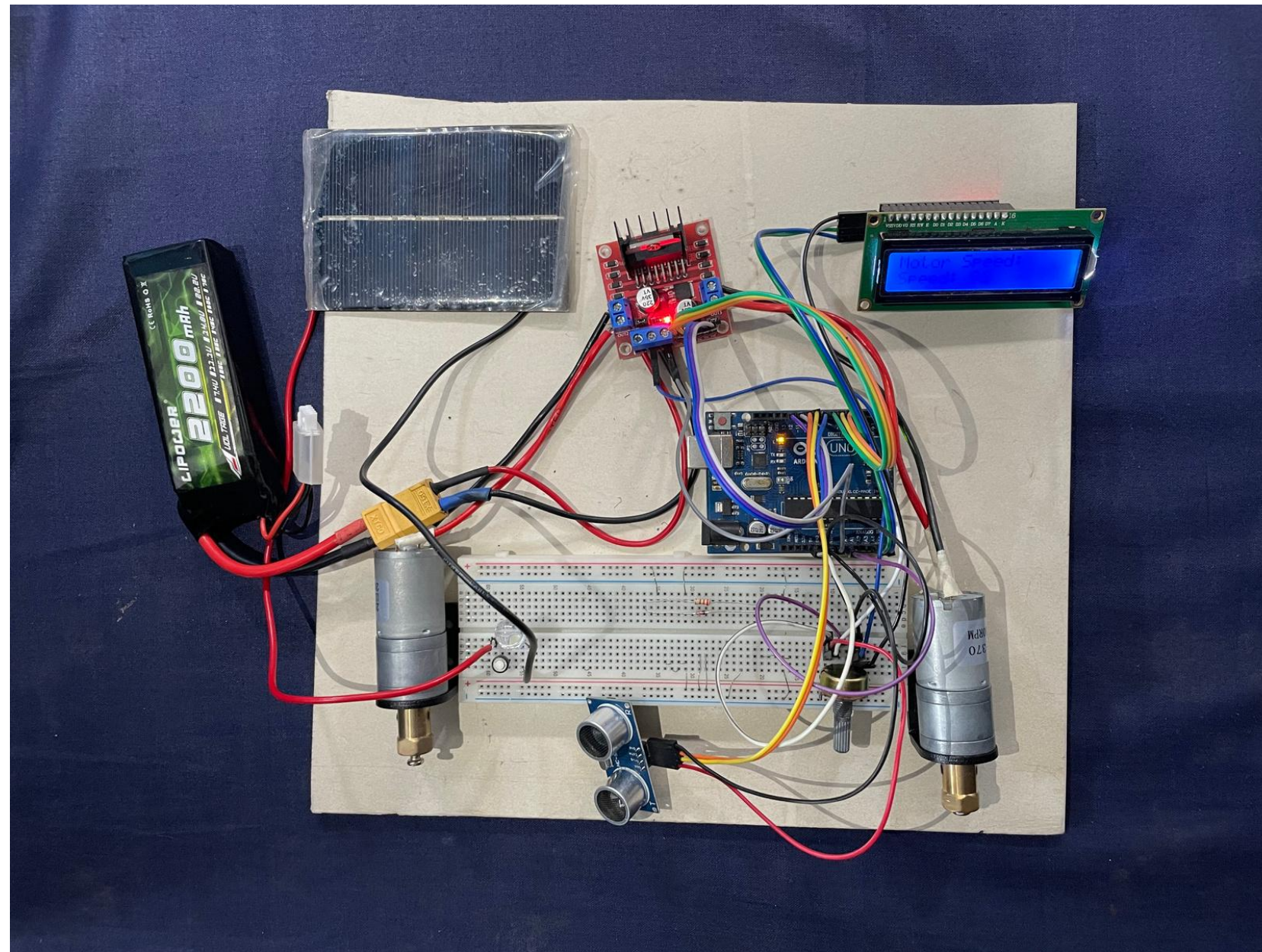
3.3 DESIGN: SIMULATION



4. IMPLEMENTATION: DEMONSTRATION



4.1 IMPLEMENTATION: PHOTO GALLERY



4.2 IMPLEMENTATION: EXTERNAL LINKS



<https://youtube.com/shorts/Rr9jDDKW8tM?si=Zxs3Y1whMCS7SmcW>



https://drive.google.com/file/d/10qWkcPVF2pdNtCMIhrEbjjUucbQbvRDR/view?usp=drive_link



<https://youtu.be/czvA3kF8Too>





5. ANALYSIS AND EVALUATION

5.1 Novelty

5.2 Project Management and Cost Analysis

5.3 Practical Considerations of the Design

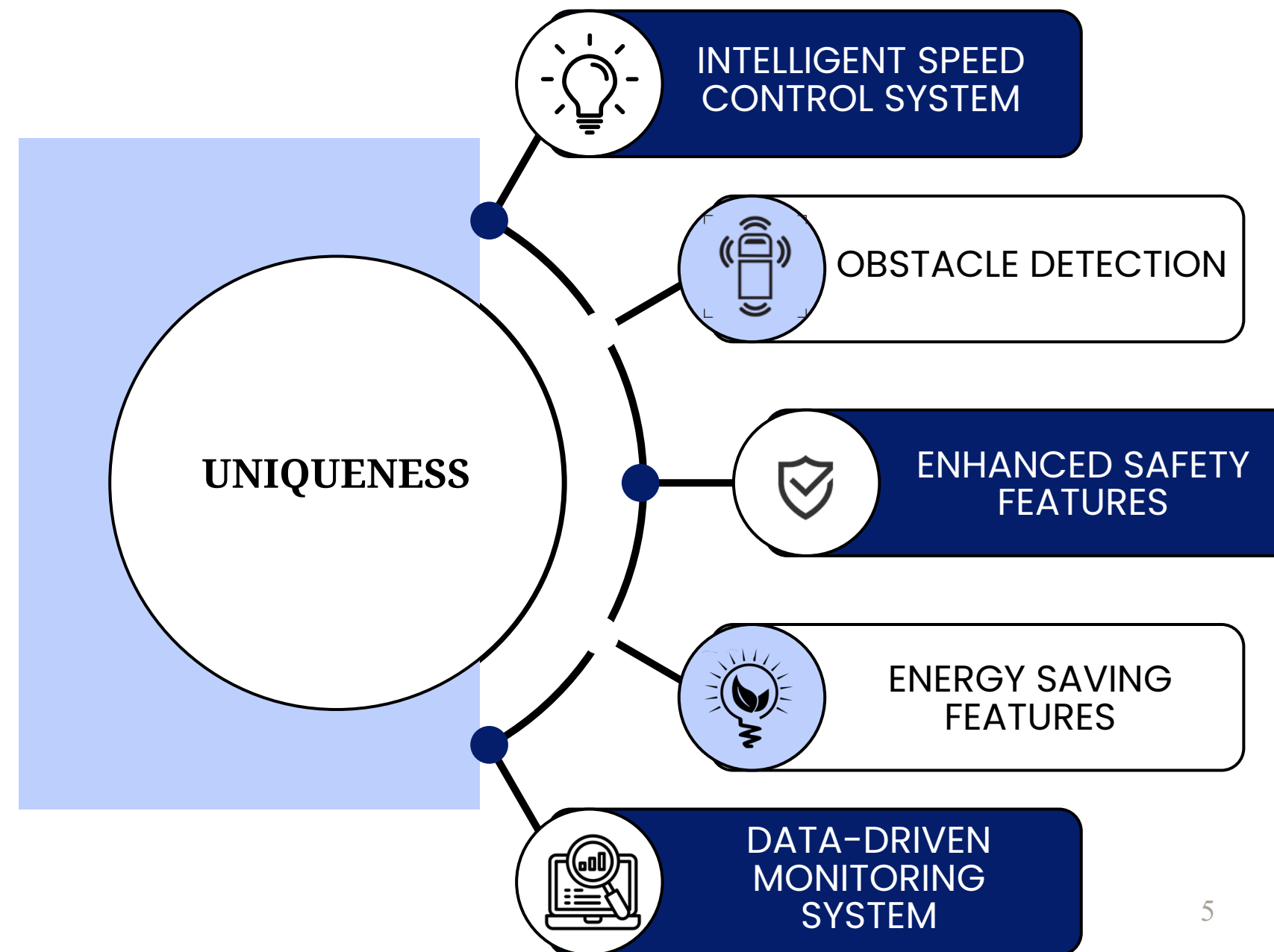
5.4 Assessment of the Impact of the Project

5.5 Evaluation of the Sustainability

5.1 NOVELTY



Our project introduces several innovative enhancements to existing battery-run rickshaws, aiming to improve safety, efficiency, and sustainability.



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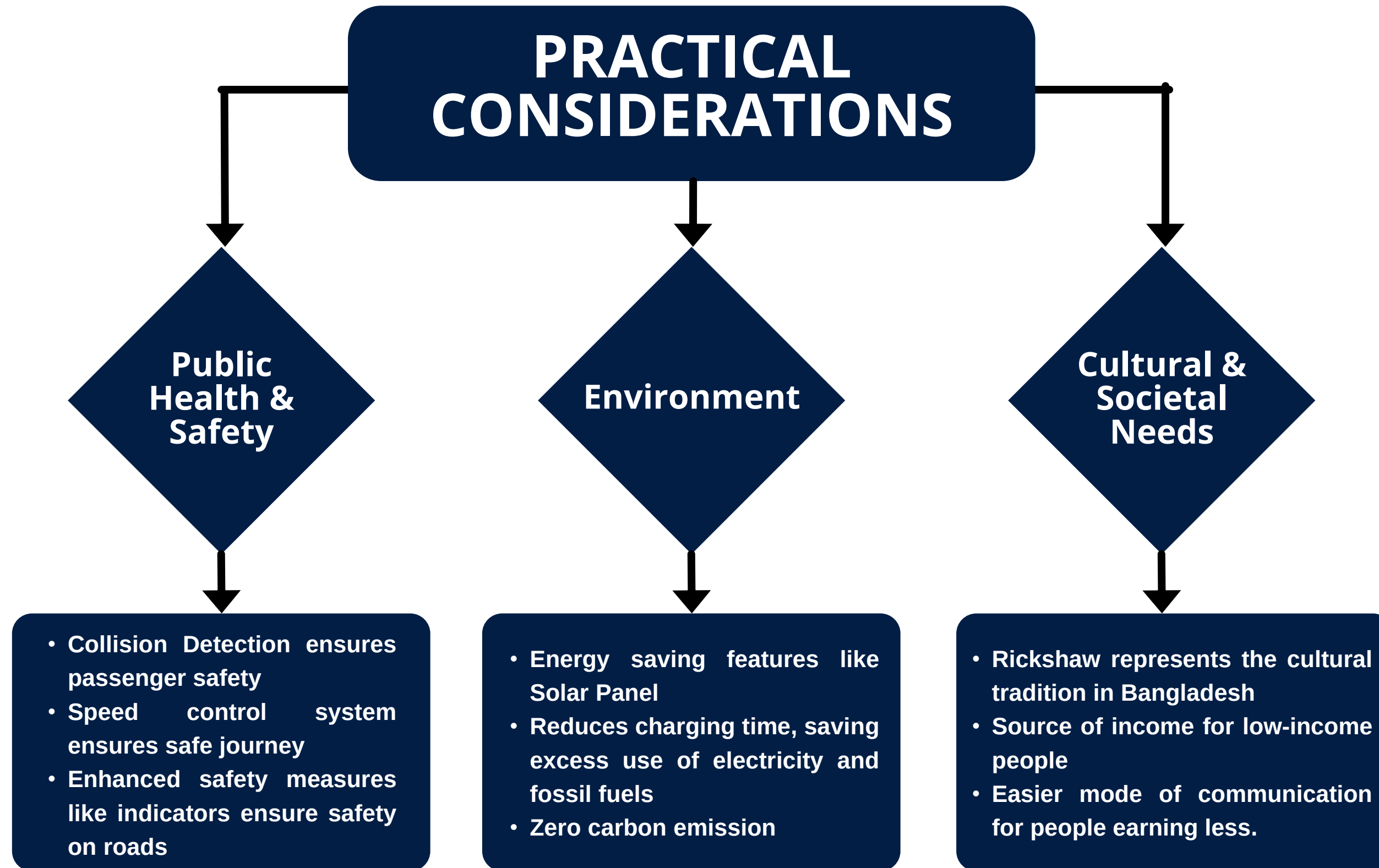
5.2 PROJECT MANAGEMENT & COST ANALYSIS



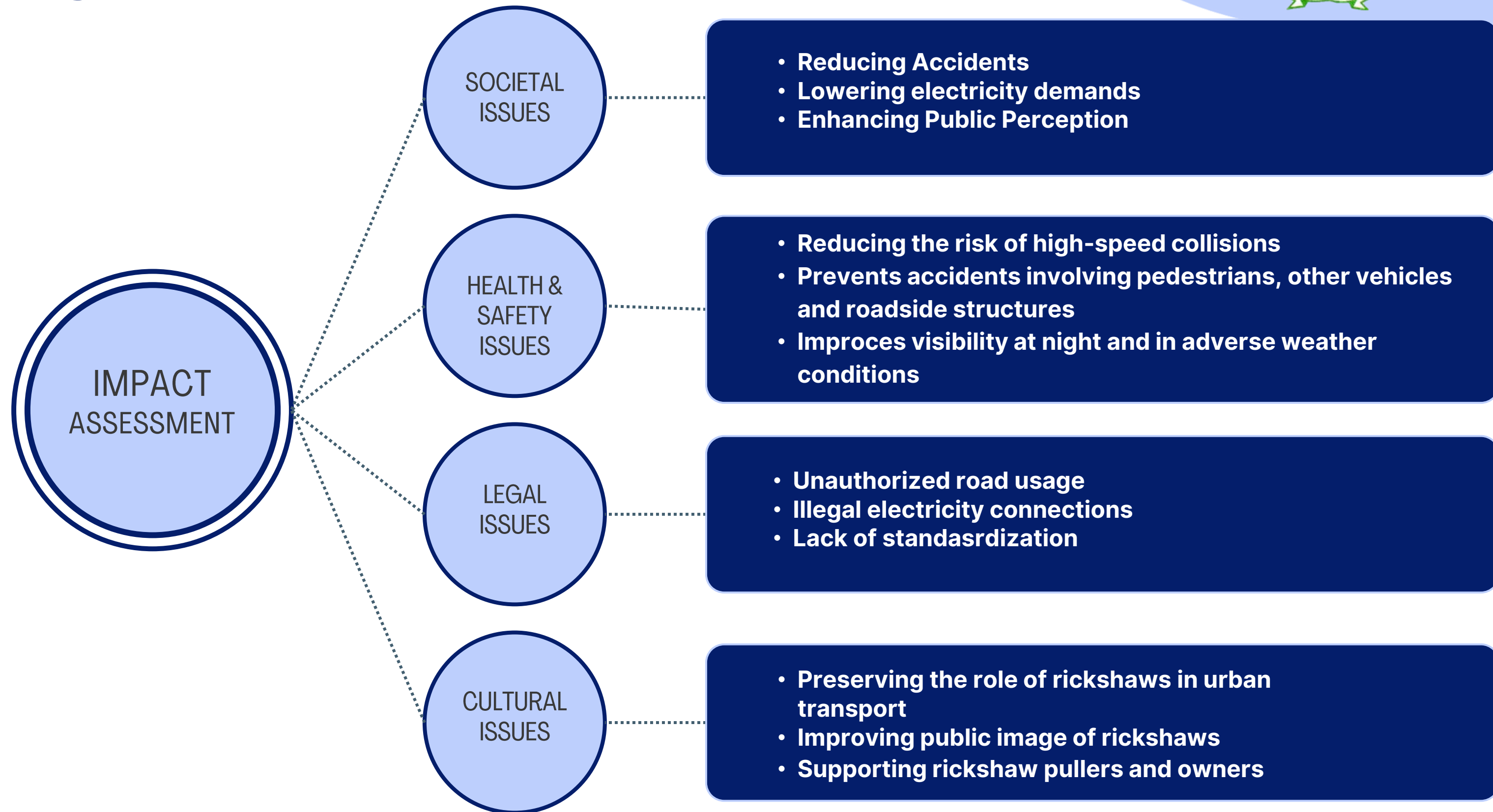
Component Name	Quantity	Price
Arduino UNO (R3)	1	995/-
Motor Driver (L298N)	1	120/-
Potentiometer (10K)	2	30/-
SONAR sensor (HC SR-04)	1	60/-
LiPo Battery 11V	1	1600/-
Motor (25GA 370 400 RPM)	2	800/-

Component Name	Quantity	Price
Solar Panel (RYO 474)	1	160/-
LED Indicators	1	10/-
Display	1	90/-
Adapter Module	1	90/-
Jumper wire + Hexa connector	1 set	260/-
Soldering Kit	1 Set	400/-

5.3 PRACTICAL CONSIDERATIONS



5.4 ASSESSMENT OF THE IMPACT



5.5 EVALUATION OF THE SUSTAINABILITY



Sustainability



- Enhanced Safety
- Job security and opportunities
- Improved Public Perception
- Inclusion and accessibility
- Reduction in traffic incidents

Societal Context



- Reduction in carbon footprint
- Decreased electronic waste
- Lower energy wastage
- Noise pollution reduction
- Sustainable urban mobility

Environmental Context



- Energy Efficiency
- Reduction in electricity Demand
- Long-term economic viability
- Durability and maintenance
- Environmental footprint



6. REFELCTION ON INDIVIDUAL AND TEAMWORK

6.1 Individual Contribution of Each Team Member

6.2 Mode of Teamwork

6.3 Diversity Statement of Team

6.4 Assessment of the Impact of the Project

6.1 INDIVIDUAL CONTRIBUTION OF EACH TEAM MEMBER



AR-RAFI ISHRAQ

- Research on component selection
- Purchasing components
- Hardware implementation (wiring, soldering)
- Code for individual Module
- Debugging



ISHRAK ANAN

- Research on component selection
- Purchasing components
- Hardware implementation (wiring, soldering)
- Debugging
- Preparing presentation



TAUFEEQ HASSAN OMIO

- Research on component selection
- Research on project feasibility and impact
- Hardware implementation (wiring, soldering)
- Report writing
- Preparing presentation



ASHFIQ UL RAHMAN

- Research on component selection
- Purchasing components
- Research on project feasibility and impact
- Hardware implementation (wiring, soldering)
- Report writing
- Debugging



TAHSIK ISLAM SAMPAD

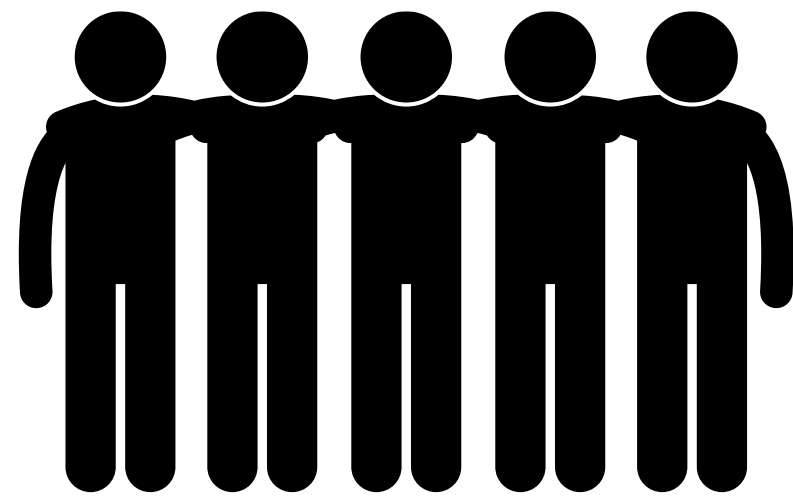
- Purchasing components
- Budget designing
- Research on project feasibility and impact
- Hardware implementation (wiring, soldering)



BAYZID MOHAMMED SAKIB

- Purchasing components
- Budget designing
- Hardware implementation (wiring, soldering)
- Documentation

6.2 MODE OF TEAMWORK



**TEAM
COLLABORATION**



**BRAINSTORMING
AND GROUP
DISCUSSIONS**



**PROTOTYPING AND
TESTING**



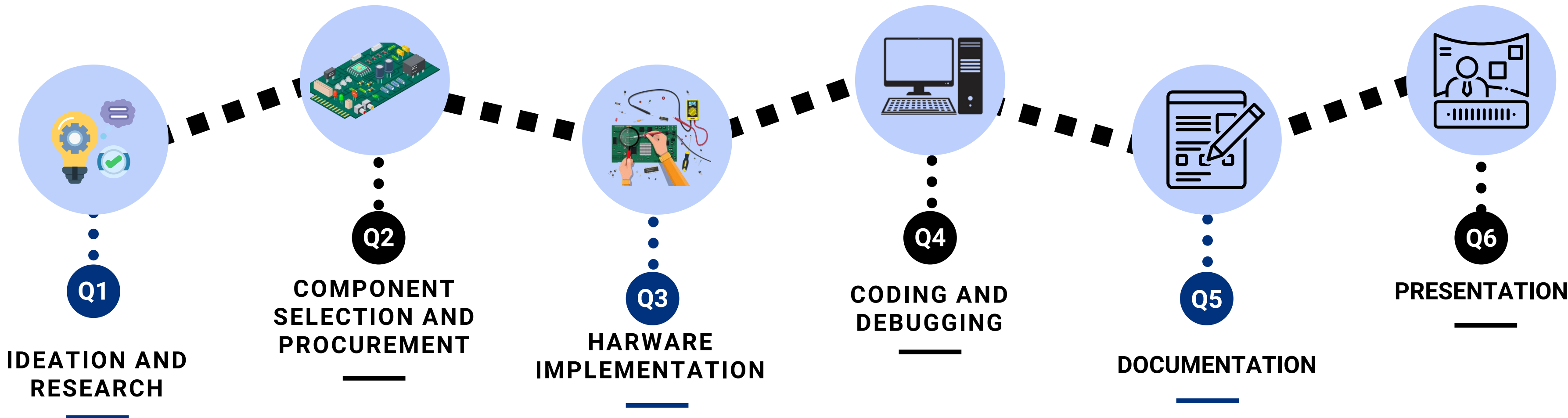
**ROLE BASED
TASK
ALLOCATION**



6.3 DIVERSITY STATEMENT OF TEAM

- **ROLE BASED TASK ALLOCATION**
- **INCLUSIVE AND COLLABORATIVE ENVIRONMENT**
- **ADAPTIBILITY AND LEARNING**
- **COMMITMENT TO ENVIRONMENTAL AND SOCIAL IMPACT**

6.4 LOG BOOK OF PROJECT IMPLEMENTATION





THANK YOU